

TEST REPORT – Matematica_LasFierbinti

1. Introduction

This report presents the testing activities carried out within the Matematica_LasFierbinti project. The purpose of the testing was to verify the main functionalities of the application, validate the behavior of the APIs and evaluate the overall quality of the product.

2. Testing objectives

- Validation of authentication functionality
- Testing the Wishlist functionality
- Testing the functionality of the shopping cart
- Testing the process of ordering a product
- Web services (API) testing
- Identifying and reporting defects
- Application security assessment according to OWASP Top 10

3. Types of Testing Performed

3.1 Manual Functional Testing

- Creation and execution of test cases in Azure DevOps
- Validation of application behavior in both positive and negative scenarios

3.2 Automated Testing

- Automation using Selenium WebDriver
- Implementation of the Page Object Model
- Automated tests executed for:
 - Valid login
 - Login invalid
 - Move product from Wishlist to Cart
 - Add valid product to Wishlist
 - Place order with empty cart
 - Place order with valid data
 - Searching with only one keyword
 - Search input persists after refresh
 - Add an item to cart as a guest
 - Add an out-of-stock item to cart

3.3 Testare API

- Verification of requests using Fiddler
- Creation of a Postman collection with assertions
- Testing the following endpoints:
 - /user/login
 - /user/logout
 - /user/createWithArray
 - / user/login?username=username&password=password
 - / store/order

3.4 Testare de securitate

- Analysis of the OWASP Top 10 security principles
- Identification of potential risks

4. Tools Used

- Azure DevOps – test case management, PBIs, bug tracking
- Visual Studio – development and execution of automated tests
- Selenium WebDriver – development and execution of automated tests
- ChromeDriver – browser execution
- Postman – API testing
- Fiddler – HTTP traffic analysis

5. Testing Results

5.1 Manual Tests

- Total test cases: 25
- Passed: 20
- Failed: 5 (reported as a bug)

5.2 Automated Tests

- Total automated tests: 10
- All tests passed after correcting the locators

5.3 Testare API

- All requests returned the correct status codes
- Postman assertions passed successfully
- Fiddler confirmed the correct structure of the requests

6. Identified Issues

Bug 1: Products can be duplicated when added multiple times to wishlist

- ID:
- Status: To do
- Reproduction: Adding the same product repeatedly to the wishlist

Bug 2: a product cannot be successfully removed from the wishlist

- ID:
- Status: To do
- Reproduction: Remove product from wishlist

Bug 3: You can place an order when delivery address is missing

- ID:
- Status: To do
- Steps to reproduce:
 - Add any product to the cart.
 - Proceed to the Checkout page.
 - Proceed to the Checkout page.
 - Click on the "Place Order" button.

Bug 4: Application fails to trigger validation for missing payment method

- ID:
- Status: To do
- Steps to reproduce:
 - Navigate to the Checkout section with items in the cart.
 - Fill in all delivery details.
 - Do not select any payment method (Card or Cash on Delivery).
 - Attempt to finalize the order.

Bug 5: Multiple consecutive clicks on "Add to Cart" creates duplicate item entries

- ID: 131
- Status: New
- Steps to reproduce:
 - Open G2A website

- Navigate to any product page
- Rapidly click "Add to Cart" button 5-6 times quickly
- Open shopping cart
- Observe cart contents

Bug 6: Adding same item to cart multiple times not incrementing quantity correctly

- ID: 132
- Status: Done
- Steps to reproduce:
 - Open G2A website
 - Navigate to any product page
 - Click "Add to cart" button
 - Navigate back to the same product page
 - Click "Add to cart" button again
 - Repeat step 4-5 two more times
 - Open shopping cart

Bug 7: Searching with special characters redirects user to products containing different kinds of currencies.

- ID: 87
- Status: New
- Steps to reproduce:
 - Go on www.g2a.com
 - Click on the search bar
 - Type random special characters into the search bar
 - Press Enter key or click the search icon

Bug 8: Searching with only character is not allowed and gives no visual feedback.

- ID: 88
- Status: New
- Steps to reproduce:
 - Go on www.g2a.com
 - Click on the search bar
 - Type a single letter/special character/number into the search bar
 - Press Enter key or click the search icon

Bug 9: No clear validation message when password field is empty

- ID: 80
- Status: New
- Steps to reproduce:
 - Go on www.g2a.com
 - Open sign in page

- Leave both fields for password and email empty
- Click the Sign in button

Bug 10: No account lockout after multiple failed login attempts

- ID: 82
- Status: Approved
- Steps to reproduce:
 - Open sign in page
 - Enter a valid email address
 - Enter an incorrect password
 - Click the Sign in button
 - Repeat steps 3 and 4 multiple times

7. OWASP Top 10 – Principles Analyzed

1. Injection
2. Broken Authentication
3. Sensitive Data Exposure
4. XML External Entities (XXE)
5. Broken Access Control
6. Security Misconfiguration
7. Cross-Site Scripting (XSS)
8. Insecure Deserialization
9. Using Components with Known Vulnerabilities
10. Insufficient Logging & Monitoring

8. Attached Artifacts

- Postman collection: *test-api-collection.json*
- Fiddler session: *fiddler-session.saz*
- Testing screenshots
- Test cases in Azure DevOps
- Automated tests (C# files0029)

